Open Set Recognition With Incremental Learning for SAR Target Classification

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***Abstract*— Synthetic aperture radar (SAR) target classification is an important application in SAR image interpretation. In practical applications, the battlefield is open and dynamic, and the SAR target classification model often encounters the targets of unknown classes. However, most of the existing SAR target classification methods follow the close-set assumption. It makes them only classify several fixed classes of targets and cannot deal with the targets from unknown classes. To this end, this article proposes a novel SAR target classification method. This method can not only classify the targets from known classes and search targets from unknown classes but also incrementally update the classification model with these unknown class targets. Specifically, an autoencoder improved by multiscale structural similarity (MS-SSIM) loss is utilized to extract targets' features, and it can better utilize the structural information in SAR images. Next, the classifier based on extreme value theorem (EVT) is established, which can classify the known class targets and search the unknown class targets. Then, we perform improved model reduction on the established classifier. This operation could speed up the model and prepare for incremental learning. Finally, after manually labeling those unknown class targets, the classifier is updated with these data in incremental form. Experimental results on the moving and stationary target automatic recognition (MSTAR) dataset indicate that, compared with the state-of-the-art methods, our proposed method has a better performance in open set recognition and incremental learning.**

***Index Terms*— Extreme value theory, incremental learning, multiscale structural similarity (MS-SSIM) loss, open set recognition, synthetic aperture radar (SAR) target recognition.**

I. Introduction

AS AN active imaging system, synthetic aperture radar (SAR) has the advantages of being all-weather, all-day, wide-range, and high resolution. Hence widely applied in various military and civil fields [1]. SAR target classification is one of the most important applications in SAR image interpretation. With the improvement of SAR image acquisition ability, a large number of SAR images have been obtained. However, compared with optical images, SAR images are more difficult to understand. Hence, it is difficult to complete the SAR target classification task in a short time only by manual interpretation. Therefore, it is urgent to develop SAR target classification technology [2].

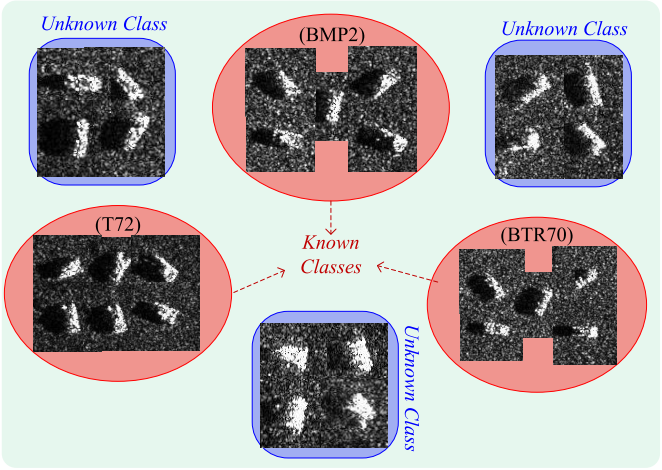


Fig. 1. Open set recognition for SAR ATR: the known class targets are BMP2, BTR70, and T72. The remaining targets are from unknown classes. The SAR ATR system should not only correctly classify the targets of known classes (BMP2, BTR70, and T72), but also determine the targets of unknown classes as "unknown."

In recent years, many SAR target classification methods have been proposed, including various traditional and deep-learning-based methods [3], [4], [5], [6], [7]. Most of them are supervised methods. Compared with manual interpretations, they have higher recognition accuracy and interpretation speed. However, there are still some obvious disadvantages, such as poor flexibility and adaptability [8]. Specifically, they usually assume that all the target classes in the test set have appeared in the training set. Hence, they cannot deal with the target classes that have not been encountered in the training set. In addition, the training and updating process of these classification models usually takes up lots of time and computing resources [9].

In practical applications, the observation environment is open and dynamic, and the number of target classes may increase over time. This places higher requirements for the SAR ATR model. In our opinion, the existing SAR ATR methods in this context can be improved in the following aspects. First, it must support open set recognition. As shown in Fig. 1, the system should not only correctly classify the targets from known classes (KCTs), but also determine the

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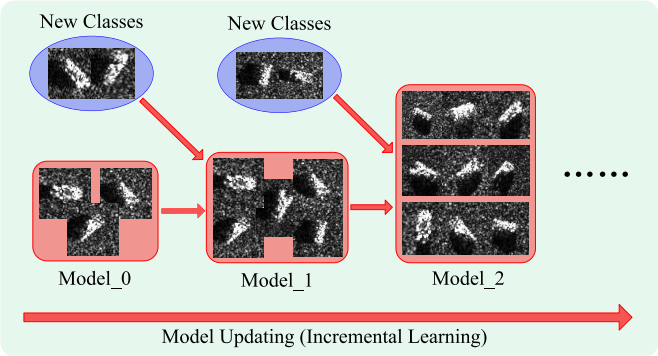


Fig. 2. Incremental learning for SAR target recognition. The model can be updated with minimal cost when it encounters the targets from new classes.

targets from unknown classes (UCTs) as "unknown." In this article, the known classes and unknown classes refer to the classes that have appeared and not appeared in the training set, respectively. In Fig. 1, the training set contains only three classes of targets: BMP2, BTR70, and T72. Therefore, the known classes are BMP2, BTR70, and T72. Other classes belong to the unknown classes. Second, it must support incremental learning. As shown in Fig. 2, after obtaining the targets from other classes, the SAR target classification system can be updated with minimal cost.

To make the SAR target recognition system more suitable for realistic scenes, scholars have carried out much work on open set recognition and incremental learning, and they have performed well. In terms of open set recognition, Sadjadi et al. [10] first applied Weibull-calibrated SVM (W-SVM) and probabilistic open set SVM (POS-SVM) to classify the KCTs and distinguish the UCTs. Both W-SVM and POS-SVM are improved SVMs, and they perform well with the Gaussian RBF kernel. Zelnio et al. [11] proposed a hybrid method to classify the KCTs and reject the UCTs, which combined convolution neural network (CNN) with SV-means (an improved support vector machine approach). Dang et al. [12] made use of the edge exemplar selection method to model the known class boundaries, and targets beyond the established boundaries were judged as UCTs. Ma et al. [13] added a classification branch to the discriminator of generative adversarial network (GAN), so the discriminator could provide the target class and the probability that the target belongs to KCTs. Then, the two kinds of information were combined to determine the target class. Hu et al. [14] improved the method in [13], so that the target recognition process could be more transparent and interpretable. Liao et al. [15] modeled each known class in SAR datasets as a multivariate Gaussian distribution in the learned feature space in EVT, and the target was regarded as UCTs if its feature was located in the low probability space of these known distributions. Inkawhich et al. [16] proposed an adversarial outlier exposure training method, and the system's ability to identify UCTs was optimized by introducing many out-of-distribution samples in the training process. Zeng et al. [17] comprehensively used the Kullback–Lebler divergence and relative position angle to process the targets' features to classify the KCTs and distinguish the UCTs at the same time. Geng et al. [47] proposed generative models via random samples combination (GvRSC). GvRSC constructs unknown targets by randomly combining the images or features from KCTs, and then the open set classifier was trained. Oveis et al. [44] extended OpenMax with various activation functions and class-wise parameter, and the extended OpenMax is more suitable for open set recognition of SAR targets. In fact, there are lots of sophisticated open set recognition methods (such as APPL [48]) that have been applied other fields, but their performance in SAR target open set recognition is mediocre.

From the perspective of incremental learning, Dang et al. [18] proposed class boundary selection-based incremental learning (CBesIL), which could save the previous recognition ability by selecting and protecting the class boundary exemplar when the classes increase continually. Dang et al. [19] improved the incremental nonnegative matrix factorization (INMF) and proposed L_p sparse constraints INMF, called L_p-INMF. Compared with INMF, L_p-INMF had better recognition performance. Tang et al. [20] proposed high plastic error correction incremental learning, which made use of the models trained for the old tasks to correct the cumulative error and reduce the model's degradation. Liu et al. [9] proposed a class-incremental SAR target recognition method, which is a deep learning method based on a continuous teacher-student learning framework. The method eased the catastrophic forgetting problem in neural networks. Guo et al. [21] proposed POSELM, and it was improved from extreme learning machine (ELM). In detail, POSELM used the particle swarm optimization algorithm to optimize the initial weight to improve the recognition performance. Tao et al. [22] proposed a multiscale incremental dictionary learning method, which is an improvement of LC-KSVD.

The abovementioned SAR target classification methods have performed well in open set recognition or incremental learning with targets from new classes. However, these two tasks are usually independent of each other. A more efficient and practical classifier should accomplish the above two tasks simultaneously. From this perspective, Dang et al. [23] proposed the open set model with incremental learning (OSmIL) method, and it performed well on the MSTAR dataset. Inkawhich et al. [49] made use of partial out-of-distribution data to help complete out-of-distribution detection. Besides, it selected part of old samples to conduct incremental learning. To our knowledge, [23] and [49] are the few works devoted to accomplish both incremental learning and open set recognition for SAR target. However, there are some drawbacks in these two methods. For example, OSmIL made use of INMF [46] to extract target features. Based on INMF-feature, an exemplar selection method was used to select edge exemplars for incremental learning. INMF is time-consuming. method in [49] require some auxiliary samples. Besides, the recognition accuracy of [23] and [49] can be further improved.

Extreme value machine (EVM) [28] is built on extreme value theory and exhibits significant performance in open set recognition. Besides, it's naturally suitable for incremental learning. In practical applications, the number of samples

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for each class needs to be specified. EVM requires multiple adjustments of the threshold using dichotomy to find the crucial samples with specific size [28], [46], which consumes a considerable amount of computational resources [46].

Motivated by the above analysis, this article proposes a new SAR target classification method for open set recognition and incremental learning. First, the autoencoder improved by multiscale structural similarity (MS-SSIM) loss is used to extract target features, which contain more structural information of SAR images. Second, the Euclidean distance between each target is calculated. Based on the distances, the boundary of each target is determined according to the extreme value theorem (EVT). Thus, the boundaries of different classes of targets can be obtained, which can complete the open set recognition task. Third, model reduction is conducted to speed up the model. At the same time, it searches the crucial samples for incremental learning. Our model reduction could directly specific the number of crucial samples, which avoids the computational burden of multiple adjustments of the threshold. Finally, we perform incremental learning with the targets of the new classes. Experimental results on the MSTAR dataset verify the effectiveness of our proposed method.

The main contributions of our article are as follows.

1) An open set recognition with incremental learning method is proposed, which is more suitable for realistic scenarios. In addition, compared with the state-of-the-art methods, our method has a better performance.
2) We introduce MS-SSIM loss to the autoencoder, which has a better ability to extract features. In addition, we establish an EVT-based open set recognition model, which is also suitable for incremental learning.
3) Generalized maximum coverage is employed to complete model reduction, which allows for the direct specification of the number of crucial samples. It avoids the computational burden of multiple threshold adjustments.

The rest of this article is organized as follows. In Section II, the related works are introduced, including open set recognition and incremental learning. Section III presents our proposed method in detail. The experimental results and related discussions are presented in Section IV. Finally, Section V summarizes the article.

II. Related Works

A. Open Set Recognition

The traditional classifiers are based on the close-set assumption, where the data in the testing set and training set share the same labels. These classifiers have achieved remarkable performance in various areas. However, realistic scenes are complicated, especially in battlefield environments. In the actual deployment of the classifiers, it is possible to encounter targets whose classes have not appeared in the training set. Obviously, the traditional classifiers cannot deal with the above problem. For convenience, the classes that have been encountered in the training set are called known classes, and the classes that have not appeared in the training set as unknown classes. An ideal classifier should not only correctly classify the targets from known classes, but also distinguish the targets from unknown classes. This is the purpose of open set recognition.

According to [24], open set recognition methods can be divided into two kinds: 1) traditional ML-based and 2) deep-neural-network-based (DNN-based) methods. Typical traditional methods are SVM-based [25], sparse representation-based [26], margin distribution-based [28], and distance-based [27] methods. The representative DNN-based methods include OpenMax [29], C2AE [30] and CROSR [31]. These methods perform well in the field of natural images. However, the performance is poor in SAR target open set recognition. To solve this problem, researchers have carried out many studies [10], [11], [12], [13], [14], [15], [16], [17]. In our opinion, these works are very meaningful for SAR ATR, but there is still room for improvement in the recognition accuracy. Therefore, we propose our method to address this issue.

B. Incremental Learning

Incremental learning uses the newly obtained data to update the existing recognition model. Compared with the traditional batch learning method, incremental learning can save considerable storage space and computing resources, and it is more efficient. Incremental learning can be roughly divided into three categories: 1) task-incremental learning; 2) domain-incremental learning; and 3) class-incremental learning [32]. We discuss only class incremental learning in this article.

Class incremental learning aims to continuously learn the new class targets at a minimum cost while not forgetting the previously learned information about other class targets. There are various class incremental learning methods, such as stored data methods [33], generative replay methods [34], regularization-based methods [35], bis-correcting methods [36], and incremental linear discriminant analysis methods [37]. The method in this article belongs to the stored data methods. These methods select a representative subset from the original training set and mix it with the data of new classes to update the model. It can significantly reduce the computational complexity. To the best of our knowledge, many scholars have focused on the class incremental learning in the field of SAR target recognition. Many works, such as [9], [18], [19], [20], [21], and [22], have achieved significant performance. However, these incremental SAR ATR methods are often conducted individually. In our opinion, before conducting incremental learning, it needs to find these new class targets from massive data. In fact, open set recognition could complete this work in a short time, which could greatly improve the processing speed. That is why we combine open set recognition with class incremental learning in the article. In what follows, we call class incremental learning as incremental learning for short.

III. Proposed Method

In this section, the proposed method is presented in detail. First, the improved autoencoder is used to extract features, which has a stronger feature extraction ability. Second, the distance between each feature is calculated. These distances

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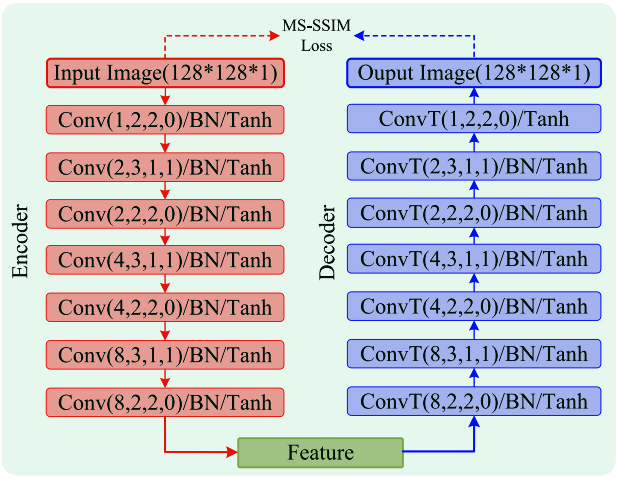


Fig. 3. Improved autoencoder for extracting features. Tanh and BN denote Tanh activation function and BatchNorm; Conv and ConvT represent convolution and deconvolution respectively. (a, b, c, d) means Out_channels $= a$, Kernel_size $= b$, Stride $= c$, and Padding_size $= d$.

are modeled following the EVT, and an open set recognition is established. Third, generalized maximum-coverage-based model reduction is conducted to reduce computation and prepare for incremental learning. Finally, after the targets of new classes are obtained, the model is updated incrementally.

A. Feature Extraction

For SAR target classification, the quality of features is crucial. Studies shows that SSIM (structural similarity) could make use of SAR image information more efficient [50], [51]. Inspired by this, we improve the autoencoder with MS-SSIM loss [38], and then it is used to extract the features. The improved autoencoder is presented in Fig. 3.

Different from the widely used mean square error (mse) loss, MS-SSIM loss pays more attention to the structural similarity of two images, rather than calculating the differences between the two images in pixels. MS-SSIM loss is an improved of SSIM loss. Hence, we first introduce SSIM loss [39]. The formula of SSIM loss is as follows:

$$\text{SSIM}(x, y) = \left[l(x, y)\right]^{\alpha} \cdot \left[c(x, y)\right]^{\beta} \cdot \left[s(x, y)\right]^{\gamma}. \tag{1}$$

In (1), x and y are the input and output images of the autoencoder; $l(x, y)$, $c(x, y)$ and $s(x, y)$ represent the brightness similarity, the contrast similarity, and the structure similarity between two images, respectively. They can be calculated according to (2). α, β, and γ are the weights to balance the three parts of loss, and they are all set to 1 in this article. In addition, μ_x (μ_y) and σ_x (σ_y) denote the average the average and standard deviation of image x (y), and σ_{xy} is the covariance of images x and y; $C_1 = (0.01 \cdot L)^2$, $C_2 = (0.03 \cdot L)^2$, and $C_3 = C_2/2$, where L is the range of gray values (L is usually 255).

Formula (2) indicates that SSIM loss takes into account the brightness, contrast, and structure similarity of the images. Therefore, the extraction feature contains more structural information of SAR images, which is very helpful for SAR target recognition. Study has indicated that applying SSIM locally in an image usually yields better results than applying SSIM globally [39]. Therefore, in practical applications, the mean-SSIM (MSSIM) is usually used to replace SSIM. $\text{MSSIM}(X, Y) = \sum_{i=1}^{M} \text{SSIM}(x_i, y_i)/M$, where x_i and y_i represent the ith image block in images X and Y

$$\begin{aligned} l(x, y) &= \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1} \\ c(x, y) &= \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2} \\ s(x, y) &= \frac{\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3} \end{aligned} \tag{2}$$

$$\text{MS-SSIM}(x, y) = \left[l_M(x, y)\right]^{\alpha_M} \cdot \prod_{i=1}^{M} \left[c_i(x, y)\right]^{\beta_i} \left[s_i(x, y)\right]^{\gamma_i}. \tag{3}$$

Studies have shown that MS-SSIM loss has better generalization ability [38]. The multiscale calculation is more adaptable to different resolution images. The image size in this article is 128 × 128, and we perform downsampling to obtain images with sizes 64 × 64, 32 × 32 and 16 × 16. The contrast and structure similarity of the images are calculated on four scales of images, and the brightness similarity is based on images with 16 × 16. Therefore, the MS-SSIM loss in this article can be calculated as above, where $M = 4$, $\alpha_M = 1$, $\beta_1 = \gamma_1 = 0.4$, and $\beta_i = \gamma_i = 0.2$ $(i \in \{2, 3, 4\})$. The values of C_1 and C_2 are referred to [52], while the rest parameters are decided according to experience.

B. Open Set Recognition

In this article, we employ the strategy in EVM [28] to establish an open set recognition model. It leverages EVT [40] to model the distances between these features, then the open set recognition model is built.

According to the EVT, for a sequence of random variables V, $\{v_1, v_2, v_3, \ldots, v_n\}$ is a batch of samples randomly taken from V, and $v_{\max}$ is the maximum of $\{v_1, v_2, v_3, \ldots, v_n\}$ $(v_{\max} = (v_1, v_2, v_3, \ldots, v_n))$. If $a_n > 0$ and $b_n \in R$

$$\lim_{z\to\infty} P\left(\frac{v_{\max} - b_n}{a_n} \le z\right) = F(z). \tag{4}$$

Meanwhile, F is a nondegenerate distribution function, and F belongs to one of the three distributions: Gumbel, Frechet, and Reversed Weibull. Among them, F obeys the reversed Weibull distribution if z is the boundary [40].

To establish the classification boundary of various targets, we use the margin distance. The margin distance of the sample x_i is defined as half of the minimum distance from x_i to all other samples belonging to different classes, and the Euclidean distance is adopted here. The margin distance of x_i can be represented by the following formula:

$$d^i = \min_{j:y_i\neq y_i} d_j^i = \min_{j:y_i\neq y_i} \frac{\left\|x_i - x_j\right\|}{2}. \tag{5}$$

In (5), y_i and y_j represent the categories of samples x_i and x_j, and $\|x_i - x_j\|$ denotes the Euclidean distance between x_i and x_j.

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It is obvious that d^i is the minimum value of the random values $d^i_j(j : y_j \neq y_i)$. If we multiply the random variables d^i_j by -1, then $z = -d^i$ is the maximum value of $-d^i_j(j : y_j \neq y_i)$, which implies that $z = -d^i$ follows the extreme value distribution. Considering that the random values $-d^i_j$ have a boundary $(-d^i_j < 0), z = -d^i$ obeys the reversed Weibull distribution. In other words, the margin distance d^i of sample x_i follows the Weibull distribution.

In this article, we use the random variables d^i_j to fit the Weibull distribution of sample x_i's margin distance, and the distribution function is presented in

$$F\left(\left\|x_i - x_j\right\|; \lambda_i, k_i\right) = 1 - \exp\left[-\left(\frac{\left\|x_i - x_j\right\|}{\lambda_i}\right)^{k_i}\right]. \quad (6)$$

In (6), k_i and λ_i are the Weibull shape and scale parameters of sample x_i.

For a sample x_t to be recognized, when the distance between x_t and x_i is smaller than the margin distance of sample x_i, x_t, and x_i are deemed to belong to the same class $(y_t = y_i)$. Therefore, the probability that x_t and x_i belong to the same class can be expressed as follows:

$$\begin{aligned}\Psi\left(\|x_i - x_t\|; \lambda_i, k_i\right) &= 1 - F\left(\|x_i - x_t\|; \lambda_i, k_i\right) \\ &= \exp\left[-\left(\frac{\|x_i - x_t\|}{\lambda_i}\right)^{k_i}\right]. \end{aligned} \quad (7)$$

In general, the process of establishing the probability model for the training set is shown in Algorithm 1. In the calculation process, a matrix operation is used to accelerate Algorithm 1.

Algorithm 1 Establishing the Probability Model for Open Set Recognition

Input:

Dataset $D = \{(x_i, y_i)\}_{i=1}^n$, where x_i and y_i denote the sample and label, n represent the number of samples in the dataset.

1: **for** each (x_i, y_i) from D **do**
2: Calculate the half-distances $d^i_j = \|x_i - x_j\|/2$, where j: $y_j \neq y_i$.
3: Form sequence $\hat{d}^i$ with τ smallest distances of d^i_j, where j: $y_j \neq y_i$.
4: Fit the Weibull distribution $\Psi(\|x_i - x\|; \lambda_i, k_i)$ with sequence $\hat{d}^i$ to get the parameters λ_i and k_i.
5: **end for**
6: **returen** $\{\Psi(\|x_i - x\|; \lambda_i, k_i)\}_{i=1}^n$

Output:

Probability function of all samples: $\{\Psi(\|x_i - x\|; \lambda_i, k_i)\}_{i=1}^n$.

Assuming that there are m samples in class C_l, we need to construct m probability functions. At the same time, the probability that x_t belongs to class C_l is $P(C_l|x_t)(P(C_l|x_t) = \max_{\{i:y_i=C_l\}}\Psi(\|x_i - x_t\|; \lambda_i, k_i))$. We think that sample x_t belongs to class C_l when $P(C_l|x_t) > \delta$, where δ is the setting threshold and $0 < \delta < 1$.

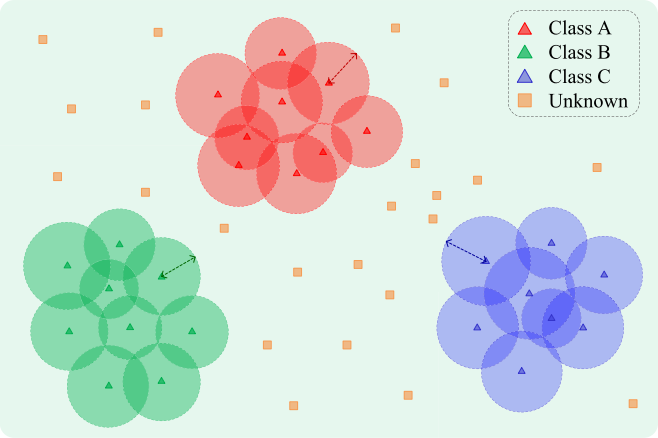


Fig. 4. Boundaries of targets from different classes.

For the multiclass recognition task, we determine the target's class according to the following formula:

$$y = \begin{cases} \underset{C_l \in \{C_1, C_2, \ldots, C_K\}}{\arg\max} P(C_l|x_t), & \text{if } P(C_l|x_t) \geq \delta \\ \text{“unknown”}, & \text{otherwise.} \end{cases} \quad (8)$$

In (8), $C_1, C_2, \ldots, C_K$ represent K different known classes, and "unknown" denotes the unknown classes.

For a sample x_i and its corresponding probability function $\Psi(\|x_i - x\|; \lambda_i, k_i)$, we can calculate the radius of its boundary when λ_i, k_i, and δ are determined. The boundaries of targets from different classes are shown in Fig. 4, and samples outside these boundaries are recognized as unknown class.

C. Model Reduction

Here we conduct model reduction for the targets of various classes, and there are two main motivations to do it. First, the trained model contains many redundant samples. Removing them could significantly reduce the computational complexity and improve the speed of target recognition during testing. Second, it is beneficial for conducting incremental learning to select some representative samples from each class of targets. It is worth noting that we perform model reduction separately for each class of targets.

The generalized maximum coverage model [54] is employed to achieve model reduction. For a certain class of samples $X = \{x_i | i = 1, 2, \ldots, N\}$, and their Weibull models are $\Psi = \{\Psi(\|x_i - x_t\|; \lambda_i, k_i) | i = 1, 2, \ldots, N\}$. It is expected to use K crucial samples (and their Weibull models) to represent the class boundary. The model coverage problem is NP-hard, and a greedy algorithm is employed to obtain a local optimal result.

Specifically, for a sample x_i, we define its cumulative inclusion probability as $f(x_i)$

$$f(x_i) = \sum_{j=1}^{|X|} \psi\left(\left\|x_i - x_j\right\|; \lambda_i, k_i\right), \quad j \neq i. \quad (9)$$

In the process of searching for crucial samples, the sample with the highest cumulative inclusion probability is selected at each iteration until K samples are selected. The generalized maximum coverage model does not take into account

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the probability of candidate samples being covered by these selected samples. Therefore, we improve it by introducing a penalty factor. As a result, the object function for each iteration becomes maximum$g(x_i) = f(x_i) - w \cdot p_i$, where $i = 1, 2, \ldots, |X|$. w is the weight factor, which could been determine through multiple experiments. p_i is the penalty factor of sample x_i, which can be represented with

$$p_i = \sum_{s=1}^{|X'|} \psi\left(\|x_s - x_i\|; \lambda_s, k_s\right). \tag{10}$$

Here, X' is the set of selected samples, and x_s is a sample in X'. In simple terms, if the sum of the inclusion probability that x_i is covered by the samples in X' is larger, it indicates that X' contains more boundary information about sample x_i. Therefore, a larger penalty is imposed when selecting candidate sample x_i. The detailed process of model reduction is presented in Algorithm 2.

Algorithm 2 Model Reduction for Selecting K Crucial Samples

Input:

Data of a certain class: $X = \{x_i | i = 1, 2, \ldots, N\}$;
Weibull models:
$\Psi = \{\Psi(\|x_i - x_t\|; \lambda_i, k_i) | i = 1, 2, \ldots, N\}$;
Number of the selected samples: K;

Initialization:

Selected samples: $X' = \phi$;
Penalty factor: $P = \{p_i = 0 | i = 1, 2, \ldots, |X|\}$;

1: **for** $k = 1, 2, \ldots, K$ **do**
2: Search the sample with maximum cumulative inclusion probability:
$index = \underset{i \in \{1,2,\ldots,|X|\}}{\arg\max} \sum_{j=1}^{|X|} \psi(\|x_i - x_j\|; \lambda_i, k_i) - w \cdot p_i$;
3: Update all penalty factors with
$p_i = p_i + \psi(\|x_{index} - x_i\|; \lambda_{index}, k_{index})$;
4: Add the selected sample x_{index} to X':
$X' = X' \cup x_{index}$;
5: Remove the selected sample x_{index} from X:
$X = X - x_{index}$;
6: Remove the penalty factor p_{index} from P:
$P = P - p_{index}$;
7: **end for**

Output:

The selected samples: X';

D. Incremental Learning

The incremental learning in this article adopts the stored-data strategy. It needs to retrain the model with the selected data during incremental learning, and the model retraining includes the establishment of open set recognition model and model reduction (as presented in Sections III-B and III-C).

Next, we will focus on the process of data selection. First, or the training dataset Data_1, an open set recognition model Model_1 is obtained. It is known that Model_1 contains some samples (Data_1′) and their corresponding Weibull models. Assuming that Model_1 has undergone model reduction, the data in Data_1′ are the crucial samples of Data_1. In other word, Data_1′ is a subset of Data_1. Second, after acquiring new data Data_new, we mix it with Data_1′ to construct the dataset Data_2 (Data_2 = Data_1′ ∪ Data_new). Finally, Data_2 was used to retrain the open set recognition model, thereby an incremental learning is completed.

The aforementioned work could complete the retraining of the open set recognition model at a lower cost, which improves the speed of model updates.

In fact, our method is improved from EVM, and the main differences between them are presented in two aspects: first, the most significant difference lies in model reduction. During the process of model reduction, EVM binarizes the sample inclusion probabilities by setting a hard threshold, then the minimum set cover method is utilized to find the minimum number of crucial samples. This results in poor control with the size of crucial samples, even though the number of samples can be adjusted by searching for the threshold using a dichotomy. However, this method incurs substantial computational and time costs. Our proposed method eliminates the need for setting threshold and enables the direct specification of the number of crucial samples, thus the computational cost associated with iterative threshold adjustment can be avoided. Second, we employ a feature extraction that utilizes an improved autoencoder, which is well-suited to the task of SAR target recognition.

IV. Experiments

In this section, we describe several experiments carried out on the MSTAR dataset [41] to verify the effectiveness of the proposed method. First, the implementations, including the MSTAR dataset and evaluation metrics, will be described in detail. Second, three-class and multiclass target recognition experiments were conducted to prove the superiority of our proposed method on open set recognition. Third, an open set with incremental learning experiment was performed. Moreover, several state-of-the-art methods were compared with our method and the results are reported in this section. Fourth, ablation studies were performed to validate the role of feature extraction and improved model reduction. Lastly, we discussed the influence of w and K on the experimental results.

A. Implementations

1) Dataset: The moving and stationary target automatic recognition (MSTAR) public dataset was collected by the air force research laboratory and defense advanced research project agency, which was imaged in HH polarization and X-band [42]. The range resolution and azimuth resolution of these images are both 0.3 m. There are ten classes of military vehicles with different depression angles and aspect angles in MSTAR data, including armored personnel carrier: BMP2, BTR70, BTR60, and BRDM2, tanks: T72 and T62, military truck: ZIL131, self-propelled howitzer: 2S1, air defense unit: ZSU234, and bulldozer: D7. As shown in Fig. 5, the SAR images of these targets are very similar, and difficult to distinguish. Therefore, it is enough to verify the performance of

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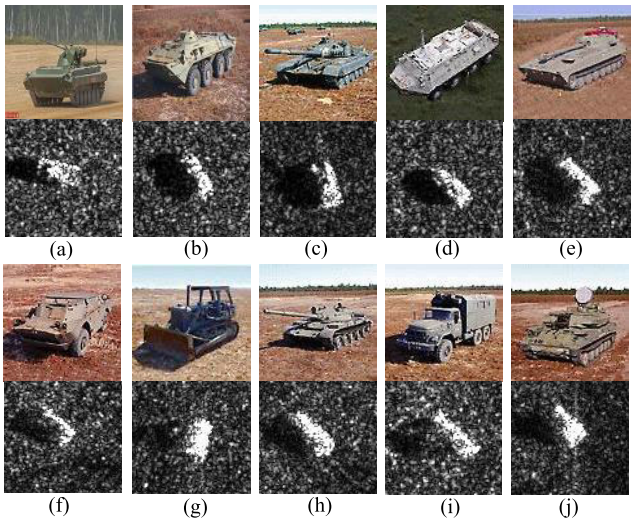


Fig. 5. Optical and SAR images of ten classes of military vehicles in MSTAR dataset. (a) BTR60. (b) BTR70. (c) T72. (d) T62. (e) 2S1. (f) BRDM2. (g) D7. (h) BMP2. (i) ZIL131. (j) ZSU234.

TABLE I
NUMBER OF TEN-CLASS MILITARY VEHICLES UNDER SOC

Class	Label	Number of Targets (17)	Number of Targets (15)
BMP2	0	233	195
BTR70	1	233	196
T72	2	233	196
BTR60	3	256	195
2S1	4	299	274
BRDM2	5	298	274
D7	6	299	274
T62	7	299	273
ZIL131	8	299	274
ZSU234	9	299	274

these open set recognition methods. The MSTAR dataset contains images under standard operation condition (SOC) [42] and extend operation condition (EOC) [43]. In this article, we select these images under SOC for experiments, in which the images with depression angles of 15 and 17 are used for training and testing the model respectively. The size of these images is 128 × 128. Moreover, the number of these ten classes of vehicles under SOC is shown in Table I.

To satisfy the needs of incremental learning, the autoencoder is trained using a training set consisting of three class targets, BMP2, BTR70, and T72. In addition, the CutMix [53] method is utilized to extend the training set. Study has demonstrated that the samples generated by CutMix can represent these unknown class samples to some extent [47], which enhances the generalization performance of the autoencoder. In the following experiments, the above-mentioned autoencoder is used for feature extraction.

2) Evaluation Metrics: First, true positive rate (TPR) and true negative rate (TNR) are used to evaluate the model. TPR and TNR can measure the model's ability to distinguish the targets from known classes and unknown classes (in fact, it is a binary classification problem), where TPR and TNR indicate the probability of correctly identifying the known class targets and unknown class targets, respectively. TPR and FPR can be calculated as follows:

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{11}$$

$$\mathrm{TNR} = \frac{\mathrm{TN}}{\mathrm{FP} + \mathrm{TN}}. \tag{12}$$

Second, we use a confusion matrix to intuitively observe the classification results. Based on the confusion matrix, accuracy, recall, precision, and $F1$-score are used to evaluate the performance of these open set recognition methods in detail. The definitions of these metrics are presented by

$$\mathrm{accuracy} = \frac{\sum_{i=1}^{K} \mathrm{TP}_{ii}}{\mathrm{Total_Num}} \tag{13}$$

$$\mathrm{recall} = \frac{\sum_{i=1}^{N} \mathrm{recall}_i}{N}, \quad \mathrm{recall}_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FN}_i} \tag{14}$$

$$\mathrm{precision} = \frac{\sum_{i=1}^{N} \mathrm{precision}_i}{N}, \quad \mathrm{precision}_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i} \tag{15}$$

$$F1 = \frac{\sum_{i=1}^{N} F1_i}{N}, \quad F1_i = \frac{2 \times \mathrm{precision}_i \times \mathrm{recall}_i}{\mathrm{precision}_i + \mathrm{recall}_i}. \tag{16}$$

In the above equations, N represents the number of target classes and Total_Num represents the number of test samples.

The experiments were implemented on a personal laptop with an Intel Core i5-9300H CPU, an NVIDIA GTX1650 GPU, and 16 GB RAM on the Windows 10 system.

TABLE II
NUMBER OF TARGETS IN TRAINING AND TEST SETS

Target Class	Known Class			Unknown Class	Total Number
	BMP2	BTR70	T72	The rest 7 classes	
Training Set	233	233	232	-	698
Testing Set	195	195	196	1838	2425

B. Three-Class Open Set Recognition

Similar to [11], [12], and [13], we selected BMP2, BTR70, and T72 to construct the training set, and the testing set contained ten classes of targets, including three known classes of targets and seven unknown classes of targets. The number of targets in the training set and testing set is shown in Table II.

In this experiment, we selected OSmIL [23], EVM [28], W-SVM with radial basis function (W-SVM RBF) [25], and one class SVM (OCSVM) [45] as the comparative methods. All of them have been applied to SAR target open set recognition. As for the features, our method utilized the features extracted by the improved autoencoder, while the other methods made use of these features extracted by INMF [19]. Additionally, both EVM and our method performed model reduction. Our method directly set the sample size to 50, while EVM adjusted the threshold several times to achieve the sample size roughly consistent with our method.

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TABLE III
1-FPR AND TPR OF SEVERAL DIFFERENT METHODS

Method	TPR (%)	1-FPR (%)
OSmIL	93.4	93.1
EVM	86.54	92.66
OCSVM	91.2	60.4
W-SVM RBF	79.7	77.6
Our Method	96.25	96.68

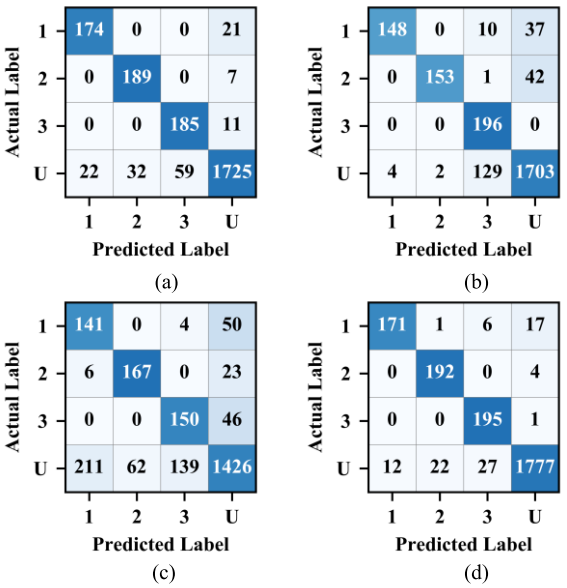


Fig. 6. Confusion matrices of (a) OSmIL, (b) EVM, (c) W-SVM RBF, and (d) our proposed method.

First, we used TPR and false positive rate (FPR) to evaluate our method's ability to distinguish targets from unknown classes, where TPR = TP/(TP + FN) and FPR = FP/(FP + TN). The results of EVM and our method were got from our own experiments, while the other results are sourced from Reference [2]. The experimental results are presented in Table III. Obviously, our proposed method exhibits the best performance, with TPR and (1-FPR) of 96.25 and 96.68, respectively. OSmIL and EVM follow closely in performance, while W-SVM and OCSVM exhibit poorer performance. Specifically, our method outperforms OSmIL by 2.85 and 3.58 in TPR and (1-FPR), respectively. These results demonstrate the superior ability of our method to distinguish the unknown class targets.

Second, to evaluate the performance of open set recognition in more detail, we compare the confusion matrices of our method with other methods, including OSmIL [23], EVM [28], and W-SVM RBF [25]. The results are presented in Fig. 6. All of these methods are traditional methods. It is clear that, compared with OSmIL, although our method exhibits slightly weaker recognition ability, it demonstrates strong recognition abilities for BTR70 and T72. Besides, our methods performed better in distinguish known and unknown class targets. Overall, it exhibits better performance in open set recognition. Concerning W-SVM, it performs significant confusion between known and unknown class targets.

TABLE IV
OVERALL PERFORMANCE OF DIFFERENT METHODS

Method	Recall	Precision	F1	Accuracy
OSmIL	93.4	82.9	87.9	93.7
EVM	86.65	87.49	85.06	90.72
W-SVM RBF	78.0	52.6	62.8	77.7
Our Method	95.44	91.97	93.44	96.29

It should be noted that, both our method and OSmIL are improved from EVM, which employs the sample-level modeling strategy, while W-SVM utilizes a class-level model strategy. In general, sample-level modeling is considered more accurate than class-level modeling. We believe that it is a contributing factor to the superior performance of our method, OSmIL and EVM.

To provide a more intuitive comparison between these methods, the overall recall, precision, $F1$-score, and accuracy were computed based on the above confusion matrices. The results are presented in Table IV. Overall, our method performed the best performance, with recall, precision, $F1$ and accuracy values of 95.44, 91.97, 93.44, and 96.29. Compared with EVM, these metrics improved by 8.79, 4.48, 8.38, and 5.57, respectively.

Compared with OSmIL, these metrics improved by 2.04, 9.07, 5.44, and 2.59, respectively. The results suggest that our method has a significant advantage over the other methods.

Through multiple experiments, we found that our method performed well when $0.01 < \delta < 0.2$. In the following experiments, we adjusted the value of threshold δ at 0.01 intervals to obtain the better results.

C. MultiClass Open Set Recognition

To further verify the effectiveness of our method in open set recognition, we construct six different scenarios to implement experiments. As shown in Table V, there are 4 to 9 known classes, and 6 to 1 unknown classes in *Scenario_A* to *Scenario_F*, respectively. This means that these scenarios have the same test set, which includes ten classes of targets. Taking *Scenario_A* as an example, its training set contains BMP2, BTR70, T72, and BTR60, while its test set includes ten classes of targets (four known classes and six unknown classes).

In the above experiment, three methods, including OSmIL, EVM, and our method, exhibited superior performance. However, since OSmIL was not released, we only compared our method with EVM in this experiment. As for W-SVM, we did not utilize it for comparison due to its poor performance on MSTAR. Furthermore, consistent with the above experiment, EVM made use of INMF features as input, while our method leveraged features extracted by the improved autoencoder. Moreover, the sample size for each class was identical to that of the above experiment.

The evaluation metrics of the two methods in various scenarios are presented in Table V. In each metric, the first column corresponds to the results of EVM, while the second column corresponds to the results of our method. Besides, $F1$ and accuracy of two methods are illustrated in Fig. 7 for a more intuitive comparison.

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TABLE V
EVALUATION MEASURES OF OUR METHOD AND EVM IN DIFFERENT SCENARIOS

Scenario	Dataset Division		Overall Performance							
	Known Classes	Unknown Classes	Recall		Precision		F1		Accuracy	
Scenario_A	0, 1, 2, 3	4, 5, 6, 7, 8, 9	83.56	94.87	85.26	91.79	82.74	93.14	88.33	95.18
Scenario_B	0, 1, 2, 3, 4	5, 6, 7, 8, 9	85.99	94.48	90.49	94.17	86.89	94.21	89.53	94.85
Scenario_C	0, 1, 2, 3, 4, 5	6, 7, 8, 9	84.44	93.14	88.49	94.72	85.10	93.68	85.32	93.44
Scenario_D	0, 1, 2, 3, 4, 5, 6	7, 8,9	82.32	92.54	88.17	93.71	83.72	92.80	81.36	91.55
Scenario_E	0, 1, 2, 3, 4, 5, 6, 7	8,9	83.93	92.65	82.43	94.08	82.02	93.11	79.51	91.63
Scenario_F	0, 1, 2, 3, 4, 5, 6, 7, 8	9	81.62	92.01	81.15	92.62	80.30	92.07	80.04	91.26

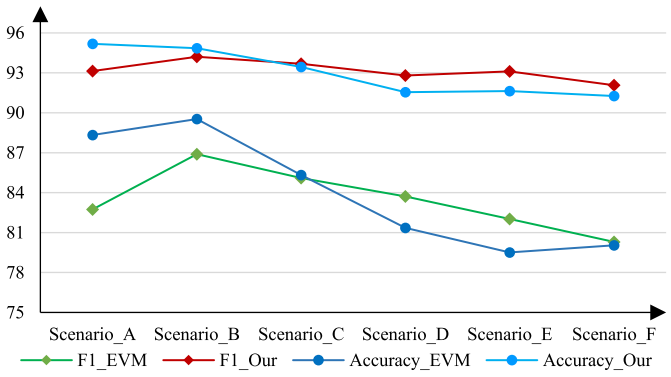


Fig. 7. Evaluation measures of our method in *Scenario_A* to *Scenario_F*.

It is evident that, our method outperforms EVM in the above six scenarios. Specifically, our method achieved the accuracy of 95.18, 94.85, 93.44, 91.55, 91.63, and 91.26 from Scenario_A to Scenario_F, which are 6.85, 5.32, 8.12, 19.19, 12.12, and 11.22 higher than the accuracy of EVM, respectively. Besides, the $F1$ scores of our method are 10.40, 7.32, 8.58, 9.08, 11.09, and 11.77 higher than those of EVM. In summary, our method exhibited excellent performance in open set recognition.

D. Open Set Recognition With Incremental Learning

Here we report the results simultaneously evaluates the method's performance in open set recognition and incremental learning. Initially, the model was employed to classify known class targets and detect targets from unknown classes. Next, the unknown class targets were amalgamated with the crucial samples from old data to construct a novel training dataset. Finally, the model was retrained on this novel dataset. thereby achieving an incremental learning process at a reduced cost. Different from conventional increment learning [9], [18], [19], [20], [21], [22], our model always maintains the ability of open set recognition. Here, we performed multiple incremental experiments.

Specifically, first, we used BMP2, BTR70, and T72 to construct the training set Set-A and train the initial SAR ATR model Model-A. At this time, the test set was composed of classes of targets: BMP2, BTR70, T72, and BTR60, which means that there was only one unknown class (BTR60) in the test set. The recognition accuracy of Model-A is shown in the first line of Table VI, where "U" indicates the unknown class and "—" means no such class. Then, we add BTR60 on the basis of Set-A, and obtained Model-B using BTR60 to update Model-A incrementally. The test set of Model-B contained five classes of targets, including four known classes (BMP2, BTR70, T72, and BTR60) and one unknown class (2S1). The recognition accuracy of Model-B is presented in the second line of Table VI. Similarly, we got models Model-C to Model-H and their respective recognition accuracy (the test set of Model-H no longer contained the targets of unknown class).

In this experiment, our method was compared with OSmIL. Similar to the previous experiments, our method and OSmIL utilized the features extracted by improved autoencoder and INMF features, respectively. Moreover, we set the sample size of each class to 80, which is comparable to that of OSmIL. The recognition accuracies of two methods are presented in Tables VI and VII. It is evident that, our method achieved the overall accuracy of 95.78, 95.56, 93.83, 92.83, 91.37, 91.96, 90.06, and 92.54 in different tasks. Compared to OSmIL, our method achieved an improvement of 2.68, 2.36, 4.93, 4.13, 4.97, 9.26, 5.16, and 6.04 in overall accuracy. Besides, as illustrated in Fig. 8, our method achieved higher $F1$ scores than OSmIL by 0.44, 0.21, 1.57, 0.78, 0.54, 2.82, and 0.26 from Set-A to Set-G. Upon further analysis of the results in Tables VI and VII, it was observed that the two methods both achieved high recognition accuracy on BTR70 and T72. For example, in Set-H, OSmIL achieved a remarkable accuracy of 95.9 on T72, which is significantly higher than the overall accuracy of 86.5. Besides, our method recorded the lower accuracy (72.99) on BRDM2, whereas OSmIL had the lower accuracy (69.9) on T62. Overall, our method demonstrated better performance than OSmIL.

In fact, both OSmIL and our method are improved from EVM. They could enable the direct specification of the size of crucial samples. It reduces the computational burden of controlling sample size in EVM. Our method exhibits significantly better performance than OSmIL, and we believe that the main reasons for this phenomenon are as follows.

1) Our method employs deep learning method to extract features, which have stronger representational ability than the INMF features used in OSmIL.
2) OSmIL selects those crucial samples prior to the establishment of the open set recognition model, while our method selects those samples after the establishment

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TABLE VI
ACCURACY OF OPEN SET INCREMENTAL LEARNING FOR SAR TARGET RECOGNITION (OUR METHOD)

	Training Set	Testing Set BMP2	BTR70	T72	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234	Overall Accuracy
	Set-A (BMP2, BTR70, T72)	99.49	98.98	96.94	87.69(U)	-	-	-	-	-	-	95.78
	Set-B (Set-A + BTR60)	94.36	96.94	97.45	94.36	87.23(U)	-	-	-	-	-	93.56
	Set-C (Set-B + 2S1)	91.28	93.88	93.88	87.69	95.99	97.81(U)	-	-	-	-	93.83
our	Set-D (Set-C + BRDM2)	95.38	99.49	100.00	97.44	95.99	90.51	77.01(U)	-	-	-	92.83
method	Set-E (Set-D + D7)	99.49	100.00	99.49	92.82	94.53	68.25	98.18	85.71(U)	-	-	91.37
	Set-F (Set-E + T62)	95.38	96.94	97.96	93.33	88.32	75.55	97.08	90.84	96.72(U)	-	91.96
	Set-G (Set-F + ZIL131)	100.00	98.47	99.49	95.38	92.34	72.63	95.99	88.28	93.80	73.72(U)	90.06
	Set-H (Set-G + ZSU234)	99.49	100.00	97.45	94.36	94.89	72.99	95.26	87.18	91.61	98.17	92.54
	Total Change	-	+1.02	+0.51	-	+1.10	-17.52	-2.92	-3.66	-2.19	-	-3.24

TABLE VII
ACCURACY OF OPEN SET INCREMENTAL LEARNING FOR SAR TARGET RECOGNITION (OSMIL METHOD)

	Training Set	Testing Set BMP2	BTR70	T72	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234	Overall Accuracy
	Set-A (BMP2, BTR70, T72)	89.2	96.4	94.4	92.3 (U)	-	-	-	-	-	-	93.1
	Set-B (Set-A + BTR60)	85.1	93.9	91.8	88.7	94.9 (U)	-	-	-	-	-	91.2
	Set-C (Set-B + 2S1)	90.8	93.4	95.4	88.2	81.0	88.3 (U)	-	-	-	-	88.9
OSmIL	Set-D (Set-C + BRDM2)	91.3	96.4	93.9	87.2	78.8	80.3	97.1 (U)	-	-	-	88.7
method	Set-E (Set-D + D7)	89.7	95.4	95.4	88.2	77.7	80.3	94.2	76.6 (U)	-	-	86.4
	Set-F (Set-E + T62)	91.3	94.4	93.4	85.1	80.3	79.2	93.8	75.5	60.6 (U)	-	82.7
	Set-G (Set-F + ZIL131)	90.3	89.1	96.9	94.9	86.7	78.5	78.5	91.2	68.1	83.6 (U)	84.9
	Set-H (Set-G + ZSU234)	92.8	96.9	95.9	89.7	79.9	79.2	93.8	69.9	90.5	85.0	86.5
	Total Change	+3.6	+0.5	+1.5	+1.0	-1.1	-1.1	-0.4	-5.6	+22.4	-	-6.6

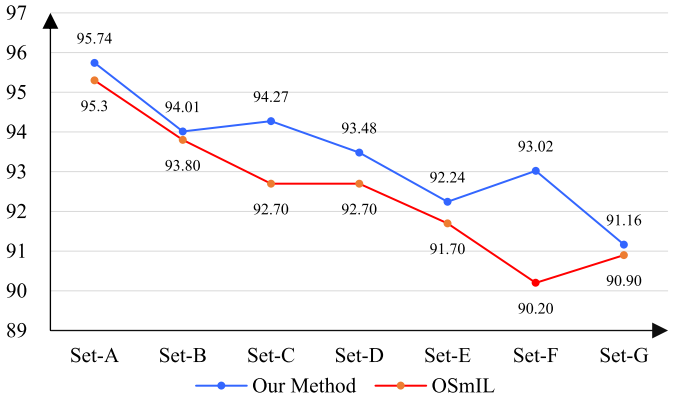


Fig. 8. $F1$ scores of our method and OSmIL.

of the model. This implies that our method utilizes more training samples during the model establishment process, which is beneficial for open set recognition.

E. Ablation Experiments

In the experiments from Sections IV-B to IV-D, as a whole, our method was compared with other state-of-the-art methods. Here we conducted ablation experiments to demonstrate the effectiveness of MS-SSIM loss and the improved model reduction. The configuration information of the methods used in this experiment is presented in Table VIII.

TABLE VIII
CONFIGURATION INFORMATION OF THREE METHODS

Method	Feature Extraction AE	AE_Improved	EVT	Model Reduction Set_Cover	Maximum_Cover
Method_1	√		√	√	
Mehtod_2		√	√	√	
Method_3		√	√		√

1) Effect of MS-SSIM Loss: In this study, Method_1 and Method_2 were employed to demonstrate the effectiveness of the improved autoencoder where AE means autoencoder with MSE loss and AE_Improved means autoencoder with MS-SSIM loss. As shown in Table VIII, the two methods both belong to EVM, but they utilized different features. In detail, Method_1 employs the features extracted by autoencoder, while Method_2 utilizes the features got from the improved autoencoder. The threshold ε was adjusted to ensure that each class of samples had the size of 50.

Following the settings in Table V, we conducted experiments in six scenarios. The experimental results are presented in Table IX. It is evident that Method_2 yields higher recognition accuracy and $F1$ than Method_1. In detail, $F1$ has been improved by 2.76, 1.45, 1.07, 1.4, 0.92, and 2.32. Accuracy has been improved by 1.28, 0.95, 0.49, 1.65, 0.86, and 0.21, respectively. The results indicates that the features extracted

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TABLE IX
EXPERIMENTAL RESULTS OF METHOD_1 AND METHOD_2

Scenario	F1		Accuracy	
	Method_1	Method_2	Method_1	Method_2
Scenario_A	90.79	93.55	93.77	95.05
Scenario_B	92.54	93.99	93.28	94.23
Scenario_C	92.35	93.42	92.54	93.03
Scenario_D	91.14	92.54	89.07	90.72
Scenario_E	92.73	93.65	91.18	92.04
Scenario_F	90.52	92.84	89.86	91.96

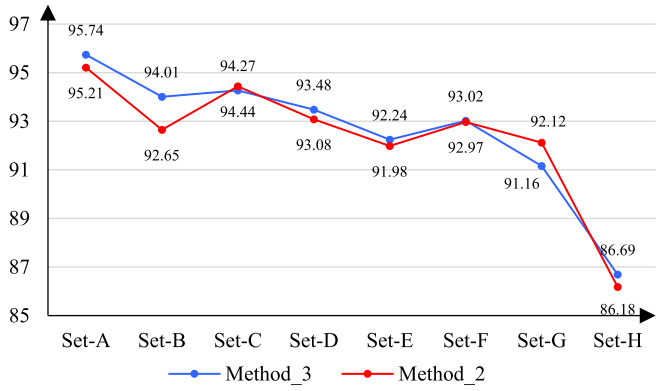


Fig. 9. Recognition accuracy of Method_2 and Method_3.

by the improved autoencoder (with MS-SSIM loss) are more conducive to open set recognition of SAR targets.

Furthermore, compared to the results of EVM (combined with INMF features) in Table V, EVM using the features extracted by autoencoder achieved better performance. For example, from Scenario_A to Scenario_F, the recognition accuracy has been improved by 5.44, 3.75, 7.22, 7.71, 11.67, and 9.82, respectively. It demonstrates that the introduction of deep-learning-based features has a significant impact on improving open set recognition performance.

2) Effect of the Improved Model Reduction: In this study, we conducted experiments using Method_2 and Method_3 to demonstrate the effectiveness of the improved model reduction. As shown in Table VIII, the two methods differ only in model reduction, with Method_2 utilizing set coverage while Method_3 using generalized maximum coverage. In fact, Method_3 is our proposed method.

Experiments were conducted following the settings in Table VI. Besides, the number of crucial samples for each class was also set to 80. The recognition accuracy and $F1$ of Method_2 and Method_3 was presented in Figs. 9 and 10, respectively.

It can be found that Method_3 has lower recognition accuracy on Set-C, Set-F, and Set-G than Method_2, but its accuracy is superior to Method_2 in the rest five cases. This conclusion is also supported by the $F1$ score. Overall, Method_2, and Method_2 exhibit comparable performance in terms of recognition results. It indicates that the improved model reduction via generalized maximum coverage has not resulted in a negative impact on the recognition results.

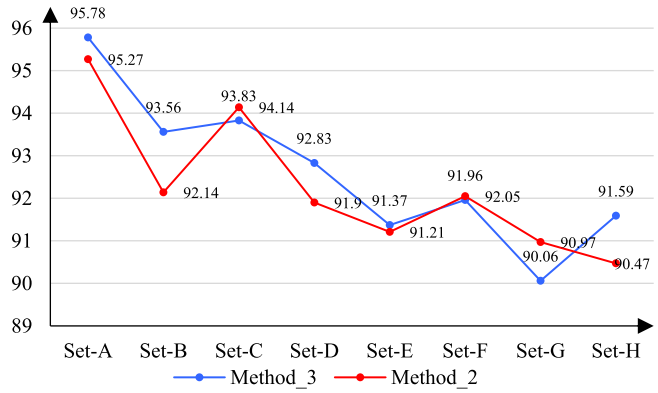


Fig. 10. $F1$ score of Method_2 and Method_3.

Furthermore, during model reduction, EVM transforms the sample inclusion probability into set cover problem by setting a certain value for the threshold ε. After that, those crucial samples are selected. EVM requires to adjust the threshold ε multiple times through dichotomy to get the expected number of crucial samples. It results in a computational complexity of $O(\log(e^{-1})N^2)$ for a single class containing N samples, where e is the termination tolerance for dichotomy. In contrast, our method and OSmIL could directly specify the number of crucial samples, with a time complexity of $O(N^2)$, Compared to EVM, our method and OSmIL are more computationally efficient, which is beneficial for incremental learning. Overall, our method is more suitable for simultaneously completing open set recognition and incremental learning tasks for SAR targets.

F. Influence of Experimental Parameters

In the above experiments, the weight factor w was set to 4 ($w = 4$) during model reduction. In addition, the number of crucial samples was set to 50 ($K = 50$) in Sections IV-B and IV-C. In this experiment, we discuss the influence of w and K on the experimental results. Taking Scenario_B of Table V as an example, which contains 5 known classes, we conduct two experiments.

First, we set the weight factor w to 0–7, respectively. Meanwhile, the number of crucial samples is set to 50 ($K = 50$). At this time, the experimental results are presented in Fig. 11. $w = 0$ means that there is no penalty factor during model reduction, which results in the worst performance with $F1 = 85.15$ and Accuracy $= 88.49$. When $0 < w < 6$, the recognition performance is constantly improved with the gradual increase of the value of w. When $w \geq 6$, the recognition results are basically stable, and the accuracy and $F1$ are basically unchanged.

Second, we set the number of crucial samples as 20, 30, 40, 50, 60, and 70 respectively. At the same time, we set $w = 4$. The experimental results are shown in Fig. 12. It is clear that, the recognition results are poor when the number of samples is small. For example, $F1 = 85.15$ and Accuracy $= 88.49$ when $K = 20$. We believe that the established class boundary is not accurate when the number of crucial samples is small. As the

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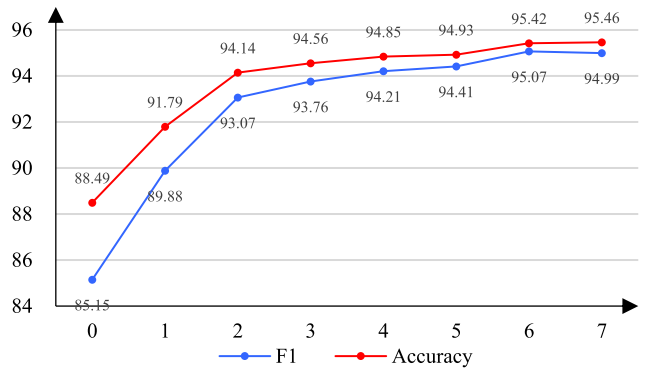


Fig. 11. Influence of weight factor w.

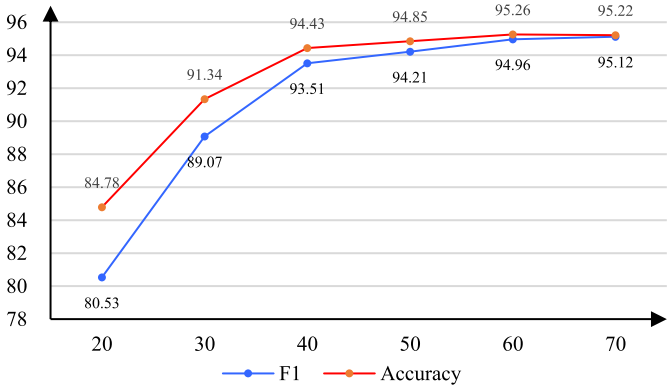


Fig. 12. Influence of sample number K.

number of crucial samples increased, the experimental results improved continuously. When $K \geq 60$, the improvement of accuracy and $F1$ becomes slight. At the same time, as the number of the crucial samples increases, the computation also increases. In the open set experiment, we set $K = 50$ to balance the recognition performance and computation.

V. Conclusion

In this article, an open set recognition with incremental learning method is proposed for SAR ATR. It has the ability to classify the known class targets and distinguish the unknown class targets. Moreover, the SAR ATR model can be updated incrementally with new class targets. This makes it more realistic for practical scenarios. Our method consists of four steps. First, the autoencoder improved by MS-SSIM loss is used to extract the features. It contains more structural information of SAR targets. Second, the distances between each target are calculated, and the open set recognition model is established by the EVT-based method using these distances. Third, model reduction is conducted to select crucial samples, which makes preparation for incremental learning and makes the model more lightweight. Finally, the model is updated in an incremental form. Experimental results on the MSTAR dataset demonstrate that our method can achieve the state-of-the-art performance in separate and integrated experiments involving open set recognition and incremental learning tasks. The effectiveness and superiority of our proposed method have been verified.

In this article, the method aims at the situation with enough training data. However, there may be few-shot problem in realistic scenarios. In future work, we will combine open set recognition with few-shot learning. Additionally, the approach that utilizing cutmix to extend the dataset may not be very suitable for the incremental learning of ship targets, due to their significant size differences. As a next step, we plan to incrementally update the autoencoder to enhance its applicability.

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